AI Builder Cup 2026 · Sustainability & Social Impact · Social Good

Second Opinion

A live scam guardian for the people we love.

Gemini listens to a suspicious call, warns in English or Hindi, brings in family, and drafts the complaint, before the money moves.

Live app: https://gen-lang-client-0095149296.web.app · Code: https://github.com/dager23/second-opinion

The problem: scams persuade, they don’t hack

¥142.3B

Japan 2025 special-fraud losses (+98%); fake-police ¥100.5B; people 65+ bore 59.2%.

₹1,935 cr

India 2024 “digital arrest” losses across 1.23 lakh complaints. Total cyber-fraud 2025: ₹22,495 cr.

A$2.18B

Australia 2025. People 65+: 17.1% of the population, 26.5% of losses.

Most money leaves through payments the victim **authorises under pressure on a call**. Banks watch transactions, but nobody listens to the call. Singapore recovered only ~15% of 2025 losses after the fact: **prevention beats recovery**.

Sources: Nippon.com, Inc42, MHA/ThePrint, NASC, Singapore Police Force (links in the app’s Impact page)

Who we protect: seniors first, then everyone

core

Seniors 65+

Digital arrest, fake bank/KYC, relative in trouble, romance.

Working adults

Investment, courier/parcel, e-commerce. 85.2% of Singapore's 2025 victims were under 65.

Young adults

Job/task scams, fake sellers, loan apps. Job scams: S$123.5M in Singapore.

Teens

Sextortion and gaming giveaways. 11.3% of Australian adolescents report sexual extortion.

One engine detects the **coercion state** of a conversation, so the same product extends to every age group by changing the audience profile and who the guardian is.

How it works

1. Listen
Speakerphone, Google Meet tab, or demo call. 16 kHz audio from the browser.

→

2. Understand
`gemini-3.8-live` transcribes and reasons about coercion signals. Calls `raise_alert` with signals + exact quotes.

→

3. Warn
Calm whisper in the user's language. Big "Hang up & call family" button.

→

4. Act
Case Agent (Gemini Flash function calling): official advisory, guardian alert, cool-off hold, complaint draft for 1930.

Frame Check
Meet video stills → visual cues (uniform, fake warrant). Cues, never a deepfake verdict.

Scam Check
Screenshot, audio recording, pasted text or a question → structured verdict, red flags, safe reply, follow-up chat.

Guardian console
Family phone gets realtime alerts and approves the hold.

Demo flow

1. A teammate plays a fake CBI officer on **Google Meet**: parcel, money laundering, “stay on video, tell no one”, “transfer to the RBI account”.
2. Signals stack up with quotes; meter → **STOP**; Hindi whisper; video cue chip.
3. Guardian’s phone buzzes → **approves the hold** → complaint draft appears.
4. A genuine bank callback triggers **nothing**.
5. Scam Check: WhatsApp task-job screenshot, a voice note, and a Teen-mode sextortion DM.

Architecture (all on Firebase + Gemini)

Frontend
React + Vite + TypeScript on **Firebase Hosting**. AudioWorklet → 16 kHz PCM.

Gemini via Firebase AI Logic
`gemini-3.8-live` (Live API, input transcription, tool calling, context compression) · `gemini-3.8-flash` (Case Agent function calling, Scam Check, Frame Check, structured JSON) · `gemini-3.5-flash-lite` (chunked fallback) · multi-speaker **Gemini TTS** for demo calls

State
Firestore (sessions, alerts, hold, report) · **Anonymous Auth** · **App Check** (reCAPTCHA v3)

Grounding & safety
Curated official-advisory library with links (I4C, RBI, Sanchar Saathi) · quoted evidence · guardian approval · no stored audio · recorded-replay fallback

Designed for the free Spark plan; the scale path moves the agent to Cloud Run + ADK (see roadmap).

Evaluation: scam *and* genuine calls

100%
of 10 scripted scam calls detected (warning or stop)

0%
false alarms on 6 genuine calls

10.0s
median estimated seconds to first warning

English, Hindi, Hinglish and Japanese scripts, run through the **same classifier prompt the app ships** (`shared/prompts.json`). Scripted, small-scale and indicative, not a field trial.

Model: gemini-3.5-flash-lite. Full table: eval/results.md in the repo.

How much could be avoided?

12%

Conservative: NAB (Australia) real-time in-app scam prompts led customers to abandon 12% of flagged payments.

up to 94%

Ceiling: a Nottingham lab experiment found confirm-the-action friction cut fraudulent payments by up to 94% (warnings alone: 18%).

Applied to published loss pools (illustrative, not a forecast): **12% of Japan's 2025 fake-police losses ≈ ¥12B a year**; 12% of India's 2024 digital-arrest losses ≈ **₹230 crore a year**. India's 1930 rapid-reporting system already helped save ~₹8,189 crore by Dec 2025, showing speed matters. Our instant complaint draft feeds that channel.

Judging criteria, one by one

Criterion	How Second Opinion answers it
Technical merit & Gen AI (40%)	Live API streaming with tool calling and context compression; Flash function-calling agent; multimodal Scam Check; structured outputs; chunked fallback; evaluation harness on genuine + scam calls.
Problem alignment & impact (25%)	Targets the largest, fastest-growing loss category for the most-targeted group; sourced stats; avoidable-loss model; feeds 1930 reporting.
Innovation & creativity (25%)	Coercion-state reasoning (not keyword matching) in Hindi/English; Meet video-cue check; guardian-approved cool-off hold; auditable quotes + official advisories.
UX & solution design (10%)	One calm screen for seniors: big type, whisper warning, one button; separate guardian phone view; works with no login; accessible, responsive.

Roadmap: from one grandparent to every age group

Teen Shield

Sextortion, gaming and giveaway scams; risk-only alerts to a trusted adult.

Young adults & merchants

Job/task scams, fake sellers, loan apps, fake payment screenshots.

Bank & telco partners

Real cool-off API, call-risk metadata, auto-reporting to Sanchar Saathi / 1930.

On-device Gemma 4

Private offline detection, Android call screening, WhatsApp bot, browser extension.

Cloud Run + ADK

Server-side multi-agent case handling, Agent Platform memory, observability.

More JAPAC languages

Gemini Live Translate for Tamil, Bahasa, Japanese, Korean and more.

Safety, privacy & feasibility

- No raw audio or video stored: only risk level, signal names, short quotes and the complaint draft.
- Guardian approves any hold; nothing is filed automatically; we show the exact words behind every alert.
- Free-tier prototype: production would use paid-tier or Vertex AI data controls, bank/telco partnerships and consent flows.
- Runs on Firebase’s free plan today; cost per protected call is a few minutes of Live audio.

Team: (add team names) · https://gen-lang-client-0095149296.web.app · https://github.com/dager23/second-opinion